Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahaj-bhadja sahaj-bhadja

Summary

Software engineer with 2+ years building Python and C++ oriented systems, API integrations, and data-processing workflows. Strong in multithreading, SQL, distributed services, debugging, and reliable delivery in fast-moving engineering environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Maintained SQL-backed services for 500+ users, tuning indexed queries, background workers, and validation flows across FastAPI workflows to improve dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant LLM job orchestration with explicit state transitions, failure isolation, and CI/CD-backed releases, processing 5,000+ monthly workflows while improving operational visibility and backend reliability under concurrent load.
- Collaborated on data fixes, release workflows, and service integrations, translating stakeholder requests into maintainable backend changes while documenting debugging paths that made recurring production issues faster to isolate.
- Supported asynchronous processing across production workflows, tracing failures through service boundaries and validating release behavior so platform updates remained stable while concurrent jobs and integrations scaled over time.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered API and ETL integrations using Python, SQL, AWS Glue, and S3, improving response time by 30% while supporting scalable service integrations, data collection workflows, and maintainable backend components.
- Maintained 20+ automated tests and CI pipelines with Jenkins, PyTest, and Mockito, increasing release reliability by 68% and strengthening debugging discipline across validation checks, deployments, and multi-environment builds.
- Worked within a microservice architecture spanning multiple operating systems, supporting executable builds, nightly validation runs, and cross-environment troubleshooting for production-style workflows with strict reliability and uptime expectations.
- Optimized API behavior for constrained environments through payload controls, response filtering, and backend tuning, improving system efficiency while preserving correctness across heavily integrated data-sharing and synchronization workflows.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Connected multimodal LLM and OCR services to backend endpoints, added concurrent batch processing, and supported CI/CD-backed releases that reduced feedback turnaround by 60% while keeping API behavior stable.
- Developed Flask APIs and batch execution workflows for an AI grading system handling 10,000+ daily requests, strengthening asynchronous processing, debugging discipline, and delivery speed across high-volume production-style operations.

Skills

Languages: Python, C++, Java, JavaScript, TypeScript, SQL, Bash
Systems & Concurrency: multithreading, background workers, distributed services, TCP/IP fundamentals, OS concepts, IPC patterns, real-time workflows
Backend & APIs: FastAPI, Flask, Node.js, Express, REST APIs, WebSockets, API integration, JSON, microservices
Data & Storage: PostgreSQL, MySQL, MongoDB, SQL Server, Azure SQL, query tuning, indexing, ETL pipelines, data modeling
Cloud & Delivery: AWS, Azure, Docker, Jenkins, Git, GitHub Actions, CI/CD, PyTest, Mockito, unit testing, debugging, Agile/Scrum

Projects

CPU Scheduling Simulator Live | *C++, React.js, EJ-2*
- Implemented FCFS, SJF, SRT, RR, priority scheduling, and MLQ with C++ logic, interactive controls, and dynamic metric calculation across 1,000+ simulated process combinations and execution scenarios.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB, Plaid*
- Built a real-time fintech platform supporting 50,000+ users and 1M+ transactions, combining secure APIs, live synchronization, and scalable infrastructure for high-volume data access and operational visibility.

Linguistix | *Python, AWS, Docker, CI/CD*
- Developed a containerized Python REST API on AWS with automated CI/CD, validation checks, and reproducible releases, achieving 87% classification accuracy while supporting reliable data-processing workflows in production-style environments.